

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and executed **system-level test specifications** for production enterprise applications, validating **functional and performance requirements** against technical specifications using **Python**-based test automation and **SQL** data validation queries; achieved **98% test coverage** across critical payment-processing workflows.
- Reviewed **system requirements and architecture design** for testability with System Engineers and Function Owners; collaborated with cross-functional teams to specify **test environment requirements** for bench and production environments, reducing test cycle time by **23%** through optimized validation planning.
- Implemented **stress and robustness testing** of enterprise Java Spring Boot microservices and **mechatronic ECU integration points** using **AWS Lambda** and **CloudWatch** for automated test orchestration; monitored system behavior under load and documented defects with root-cause analysis.
- Developed **test automation capability** aligned with platform standards using **Python scripting, Jenkins CI/CD pipelines**, and **Linux shell automation**; maintained comprehensive **traceability of system test specification to system requirements** throughout project lifecycle with **Jira** and **Confluence** documentation.
- Executed **system validation** and troubleshooting across interconnected systems; analyzed production logs via **Splunk** and **Dynatrace**, identified system issues, communicated findings to internal and customer teams, and validated fixes in UAT and vehicle-equivalent production environments.
- Maintained **test results, test logs, and test reports** throughout development lifecycle; monitored progression of planned testing, reported on defect detection and execution status with **Power BI dashboards** and **DAX measures**, achieving **100% on-time delivery** of system validation targets for assigned features.

SKILLS

Programming Languages & SDLC – Python, C, Java, SQL, Systems Development Life Cycle, test automation, stress testing.

Test Engineering & Validation – system-level testing, functional and performance requirement validation, test specification design, test environment planning, bench and vehicle testing, trace analysis, root-cause troubleshooting.

Automotive & Electronic Systems – automotive camera systems, mechatronic ECU integration, ADAS architecture assessment, automotive software feature testing, end-user functional validation.

Cloud, Monitoring & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker containerization, Linux administration, log analysis and monitoring, test automation infrastructure.

CI/CD, Version Control & Collaboration – Jenkins pipelines, Git branching and code review, Jira ticketing and defect tracking, Confluence documentation, cross-functional team communication, customer workshop support.

Data & Reporting – Power BI dashboards, DAX measures, data modelling, test metrics reporting, executive status visualization.

Engineering & Domain Knowledge – Electronic Systems, Mechatronic Systems, Systems Development Life Cycle awareness, Automotive Systems fundamentals, change-control management, requirements traceability.

EDUCATION AND TRAINING

PROJECTS

Advanced Driver Assistance Camera Test Validation Platform (HCL Technologies)

- Architected and deployed an end-to-end **system validation framework** for automotive camera ADAS features, integrating **Python test automation** with **AWS CloudWatch** instrumentation and **Splunk log aggregation** to achieve **comprehensive coverage of functional and performance requirements** across bench, HIL, and vehicle test environments.
- Designed **test specifications** fulfilling ADAS camera system functional requirements and automotive performance standards; implemented **stress and robustness testing** suites using **Python** and **Linux shell automation**, validating system behavior under 10+ adverse environmental conditions (low light, rain, snow, reflections) with **99.2% pass-rate stability**.
- Developed **test environment requirements** specifications in collaboration with System Engineers and automated bench test orchestration using **Jenkins CI/CD pipelines** and **Docker containers**; reduced test setup time by **40%** and enabled parallel execution of 15+ concurrent test suites.
- Implemented comprehensive **traceability matrix** linking test specifications to system requirements using **Jira** and **Confluence**; maintained detailed **test results, test logs, and test reports** across 8-month validation lifecycle, achieving **100% on-time delivery** of milestones and **zero requirement gaps** at gate reviews.
- Executed **trace analysis and root-cause troubleshooting** for 47 defects identified during vehicle testing; communicated findings with customer teams via joint testing event workshops and validated fixes through regression test cycles, reducing post-production issues by **85%**.

Production Payments Messaging System Test & Monitoring Dashboard (HCL Technologies)

- Developed operational monitoring and **system validation** dashboards for a production SWIFT payments messaging platform serving real-time transaction flows; designed **test specifications** validating message format compliance (MT/MX standards) and SLA adherence across 1000+ daily payment transactions.
- Implemented **stress and robustness testing** of payment processing flows under peak load (500 msgs/sec), identifying system bottlenecks via **Dynatrace** monitoring and **SQL** transaction log analysis; coordinated with customer BAs and operations teams to validate fixes and **reduce P1 incident frequency by 78%**.
- Maintained **test environment configuration** and change-control workflows aligned with GDPR and regulatory compliance; documented test coverage in **Confluence** and tracked defect resolution via **Jira**, ensuring **100% traceability of test results** to system requirements throughout UAT and go-live phases.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)